VisuoGrasp: Deep Reinforcement Learning for Multi-Finger Grasping with Vision-Confirmed Contact

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Abstract

This work implements a MuJoCo grasping system with a 16-degree-of-freedom four-finger hand, two RGB-D viewpoints, a temporal contact detector, and Soft Actor-Critic (SAC). We study whether substituting camera-confirmed fingertip contact for privileged simulator contact affects learning and generalization. An unstable one-step SAC implementation was diagnosed through rapidly increasing critic values; a revised LayerNorm and ten-step-return implementation enabled twelve complete training runs across three seeds and four experimental conditions. In an independent evaluation, vision-based SAC achieved **62.6% ground-truth grasp success** on the six training object families and **70.3%** on five unseen families, compared with 43.9% and 57.0% for a random-action policy. However, the vision-based policy remained below the scripted Level-A controller (86.7% nominal), and its visual confirmation success was lower than its physical success. At Level B, SAC with a scripted prior reached 50.9% nominal success, versus 87.2% for a hand-designed steering controller. The results demonstrate learnable multi-finger behavior and improved training stability, while identifying limitations in visual contact feedback, generalization, and wrist control.

1 Introduction

Multi-finger grasping requires coordinated control of many joints, stable contact with an object, and the ability to recover when contact is uncertain. Conventional simulated reinforcement-learning agents often access exact contact forces or contact flags from the physics engine. Such privileged signals are difficult to obtain on a physical robot. The VisuoGrasp project instead asks whether a camera-based detector can confirm fingertip contact well enough to drive an RL controller.

The system follows the provided VisuoGrasp specification: a four-finger hand, procedural and held-out object families, two cameras, contact-confirmation logic, and a physically defined success test. We focus on three questions: (i) can an agent learn without scripted finger closure, (ii) what is the performance cost of camera-confirmed contact relative to exact simulator contact, and (iii) can learned actions generalize across object geometry and initialization changes?

A major part of this work was identifying why early policies were unstable. Consequently, we present both the final results and the diagnostic progression from the initial physics and critic failures. The objective is a *reproducible comparison*, not the highest training-set success observed at a single checkpoint.

2 Simulation Environment and MDP

2.1 Hand, objects, control levels

The environment is implemented in MuJoCo [1]. The hand has four fingers with four independently actuated joints each, giving 16 finger degrees of freedom. Level A contains 16 policy actions; the robot starts close to the object, and a scheduled lift follows finger closing. The agent must learn finger closure itself, without a scripted closure prior. Level B adds three translational wrist actions (19 actions total). In the evaluated Level-B variant, scripted descent, closure, and lift form a prior, while SAC controls wrist corrections and finger residuals. **This Level-B experiment is not the fully autonomous Level B requested by the original specification.**

Six procedural families are used in training: sphere, box, cylinder, ellipsoid, capsule, and torus-shaped ring. Test-1 changes dimensions and aspect ratios of the training families. Test-2 introduces five unseen families: cone, hexagonal prism, L-shape, mug-like object, and bottle-like object. Object size, mass, friction, yaw, and the Level-B initial hand offset are randomized; the initial wrist offset is up to $\pm 3\,\text{cm}$ per horizontal axis. The trained runs use an elliptic contact-cone solver with `impratio=10` and `noslip_iterations=5`. This physically motivated numerical stabilization is reported as part of the simulator configuration, not as proof of sim-to-real validity.

2.2 Observations and success condition

The low-dimensional observation combines joint positions and velocities, wrist state, estimated fingertip positions, an object center inferred through the perception pipeline, previous action, and contact information. V3 additionally includes commanded actuator targets and normalized episode phase. Level A has an observation vector of length 98 and Level B of length 101. In the M2 condition, the contact channels contain temporally confirmed D2 contact flags and detector probabilities. The Blind ablation runs the same D2 detector and receives the same CV-based training reward but hides contact flags and probabilities from the actor. M0 instead uses simulator contact as a privileged reference.

Let h_t be the object's displacement above its resting height and $c_{i,t}$ denote physical pad contact for finger i, where $i = 0$ identifies the thumb. The evaluation success condition is

$$\text{GT success} = 1 \text{ if } \quad h_t \geq 0.10\,\text{m for at least } 3\,\text{s}, \qquad c_{0,t} = 1, \quad \sum_{i=1}^{3} c_{i,t} \geq 1. \tag{1}$$

CV success uses the same height-and-hold condition but substitutes CV-confirmed contacts. The physical ground truth is used for independent scoring, detector labels, and the M0 comparator; it must not be substituted for the live contact signal in M2.

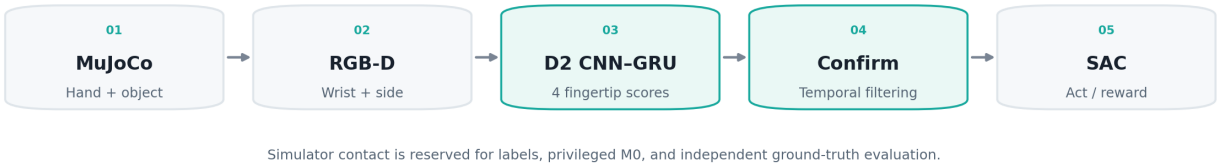


Figure 1: VisuoGrasp perception-to-control loop. The detector and temporal confirmation supply the contact signal in the vision-based M2 condition.

3 Visual Contact Detector

Two RGB-D viewpoints, one wrist-mounted and one external, observe the hand and object. Three detector families were considered: D0, a geometric depth-gap rule; D1, a learned convolutional patch classifier; and D2, a temporal convolutional/recurrent model. The final D2 variant processes dual-camera features over a four-frame temporal window. A confirmation rule filters individual probabilities to avoid treating a single noisy frame as reliable contact.

Previous detector experiments reported a D1 V2 independent-test F1 of 0.842 and a D2 V2 independent-test F1 of 0.854. The same D2 model obtained F1=0.911 in a separate in-simulator benchmark. **These numbers are not interchangeable**: the independent test and the in-simulator evaluation use different sampling protocols. The temporal detector improved on the independent D1 comparator, but the suggested unseen-shape F1 target of about 0.90 was not established by that independent test.

The detector's contribution is assessed both at the frame level and at the complete-episode level. High frame-level recall during lifting does not guarantee a confirmed grip, since the success rule requires temporal persistence and simultaneous thumb-and-finger contact. In the final experiment, a physically successful episode could therefore be missed by CV confirmation.

4 SAC Architecture and Training Stabilization

4.1 Continuous-action learning

Soft Actor-Critic [2] was selected for continuous finger and wrist actions. The policy learns a stochastic distribution $\pi_\phi(a \mid s)$ while two critics estimate discounted soft returns. The actor objective is

$$J_\pi(\phi) = \mathbb{E}_{s,a\sim\pi_\phi}\left[\alpha \log \pi_\phi(a \mid s) - \min_{i\in\{1,2\}} Q_{\theta_i}(s,a)\right], \tag{2}$$

where α controls entropy regularization. Both actor and critic use two hidden fully connected layers of 256 units; the final V3.3 critic adds LayerNorm after hidden linear layers. The discount factor is $\gamma = 0.99$, batch size 256, and learning rate 3×10^{-4}. An off-policy replay buffer and target critics are used during learning.

The shared reward favors geometric approach, lifted height, CV-confirmed contact, a thumb-opposition pair, sustained holding, and penalizes dropping and excessive motion. Potential-based geometric shaping is combined with explicit contact and hold terms. The M0 run substitutes privileged contact for the CV signal, while M2 and Blind preserve CV-based reward terms.

4.2 Diagnosing unstable values

The initial V3.1 SAC policy deteriorated after reaching useful intermediate checkpoints. In one Level-A M2 run, validation GT success fell from 38.9% at 50k steps to 11.1% at 200k. Diagnostic logs revealed an associated rise in estimated Q-values to tens of thousands and critic loss to millions. V3.2 introduced LayerNorm and observation clipping, but Q continued rising beyond 59 by 20k in a Level-B run. The next revision retained these safeguards and used ten-step returns.

For $n = 10$, the learning target takes the form

$$y_t = \sum_{k=0}^{n-1} \gamma^k r_{t+k} + \gamma^n (1 - d_{t:t+n}) \left[\min_i Q_{\bar{\theta}_i}(s_{t+n}, a') - \alpha \log \pi(a' \mid s_{t+n}) \right], \tag{3}$$

with termination handling in the replay implementation. The final training version combined ten-step replay with LayerNorm; these components were *not* isolated in the full three-seed study, so we do not attribute all improvement causally to a single component. In V3.3, mean logged Q-values on Level A remained roughly 25–40 across 100k steps, rather than showing the earlier runaway growth.



Figure 2: Mean logged critic values during V3.3 Level-A training. Lines show the mean of three seeds, while bands cover their minimum and maximum. Q calibration is distinct from grasp success.

5 Experimental Design

Four configurations were trained with three random seeds $\{11, 22, 33\}$: **A-M2** (camera-confirmed contact), **A-M0** (simulator-contact oracle), **A-Blind** (CV-based reward without actor contact input), and **B-M2** (camera-confirmed contact with scripted motion prior). Each of the 12 runs had a budget of 100,000 environment steps, totaling 1.2 million steps. Evaluation occurred every 25,000 steps on 36 fixed selection episodes; the saved checkpoint was selected from these measurements. Results from this selection set were not used as final test results.

Final evaluation used deterministic actions and disjoint episode indices. For each learned configuration, the nominal benchmark included 180 episodes per seed (540 total); Test-1 included 60 per seed (180 total), and Test-2 included 100 per seed (300 total). Baselines were evaluated separately on the corresponding object sets: random actions, zero action, open-loop scripted closure, and a hand-designed Level-B steering controller. The training evaluation and final test therefore differ in both episode count and purpose.

Setting	**Value**
Simulator / hand	MuJoCo 3.17, four-finger 16-DoF hand
Policy / steps	SAC, 100,000 steps per seed, 3 seeds
Critic / replay	Twin critics, LayerNorm, ten-step replay
Learning rate / batch	3×10^{-4} / 256
Evaluation intervals	25,000 steps; 36 episodes per selection checkpoint
Independent nominal / Test-1 / Test-2	180 / 60 / 100 episodes *per seed*
Action dimensions / observation dimensions	Level A: 16 / 98; Level B: 19 / 101
Contact modes	M0 oracle, M2 temporal D2, Blind contact-input ablation

Table 1: Final V3.3 training and evaluation configuration.

6 Learning Performance

Figure 3 displays the intermediate GT success estimates during training. On Level A, the three configurations learned meaningful behavior, but the curves oscillated substantially between checkpoints. For example, the saved Level-A M2 checkpoints selected during training had validation GT success of 63.9%, 69.4%, and 75.0% across the three seeds; the third seed passed 70% on the *selection* episodes. The selected checkpoint for seed 11 occurred at 25k even though a slightly higher single observed score occurred at 75k, because the checkpoint rule required a minimum improvement of two successes out of 36.

For Level B, selected validation GT success was 38.9%, 61.1%, and 55.6% across seeds. In comparison with the finger-only condition, wrist residuals and nonzero initial offsets remained difficult. The learning curves do not establish success against scripted baselines; only the independent final evaluation can support that comparison.

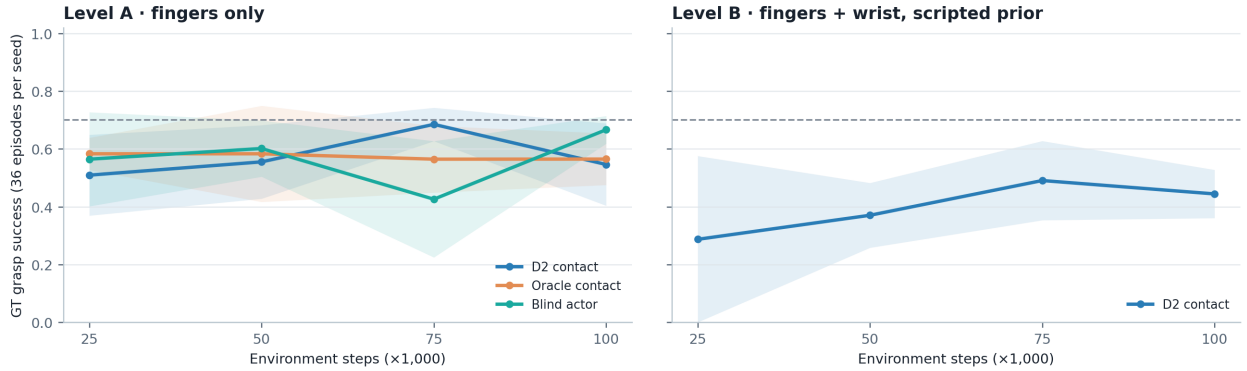


Figure 3: Training-time learning diagnostics from the final V3.3 evaluation pipeline. These are checkpoint-selection results, *not* independent test scores.

7 Independent Evaluation and Generalization

7.1 Physical grasp success

Table 2 and Figure 4 summarize the independent measurements. The mean here is calculated across the three training seeds for learned methods. On nominal Level A, SAC M2 obtained 62.6% success, compared with 43.9% for random actions and 86.7% for a scripted controller. The absolute difference against random is +18.7 percentage points; the deficit against scripted closure is −24.1 percentage points. M0 and Blind achieved 65.2% and 65.6%, respectively. These differences among learned contact variants were small relative to the variation between three training seeds.

Level	Policy	Nominal	Test-1	Test-2	Nominal episodes
A	Random actions	43.9%	38.3%	57.0%	180
A	Open-loop scripted	**86.7%**	88.3%	94.0%	180
A	SAC M2 (D2 vision)	62.6%	46.1%	70.3%	540
A	SAC M0 (oracle)	65.2%	61.1%	80.3%	540
A	SAC Blind	65.6%	61.1%	77.7%	540
B	Zero residual / open-loop	52.8%	50.0%	77.0%	180
B	Steering script	**87.2%**	81.7%	97.0%	180
B	SAC M2 (D2 vision)	50.9%	37.8%	68.7%	540

Table 2: Independent GT success, with baseline results evaluated separately. Test-1 modifies size/aspect ratio; Test-2 contains the five unseen procedural families, *not* YCB objects.

The unseen Test-2 success rate of Level-A M2 (70.3%) exceeds its nominal rate (62.6%). This is *not* evidence that unseen shapes are generally easier or that the agent has surpassed its training distribution: the geometric families and difficulty composition differ between test sets. Test-1 was more difficult for M2, with success falling to 46.1%. The result indicates sensitivity to size/aspect-ratio shifts rather than a monotonic generalization gap.

At Level B, M2 reached 50.9% nominal success, below both the zero-residual scripted prior (52.8%) and the steering controller (87.2%). Robustness plots show that SAC can sometimes preserve moderate success at large hand-object offsets where the open-loop script fails, but it did not outperform the well-tuned steering controller overall. Therefore, this experiment does not substantiate a claim that Level-B RL improves the reference controller.

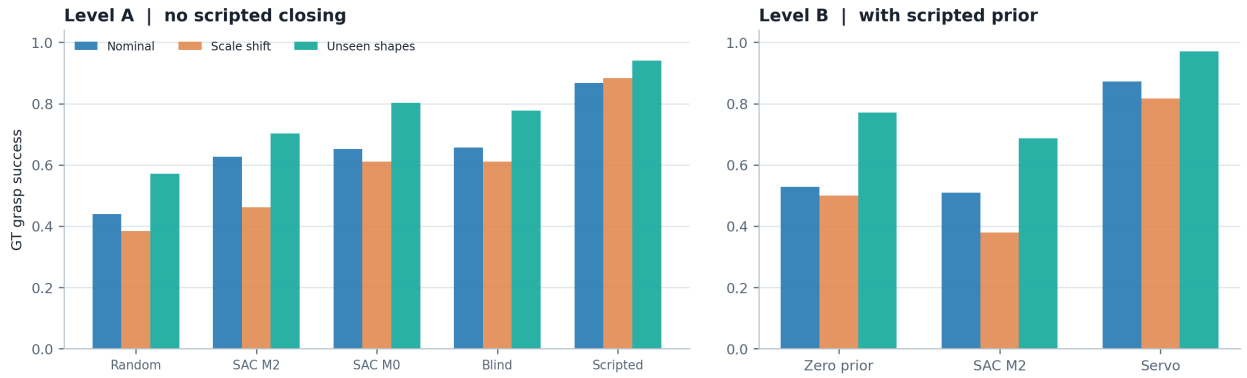


Figure 4: Independent grasping performance across the evaluation conditions. The final SAC policies improve over random actions at Level A but do not outperform the stronger scripted controllers.

7.2 GT versus CV-confirmed success

The main scientific distinction is between *physical* grasp and *visually confirmed* grasp. Level-A M2 achieved nominal GT success of 62.6% but CV success of 56.7%. On Test-1, the corresponding values were 46.1% and 35.6%; on Test-2 they were 70.3% and 50.0%. Thus, the GT–CV gap grew to 20.3 percentage points on unseen shapes. This is compatible with visual false negatives under policy-generated configurations, but a single episode-level gap alone cannot identify occlusion, misclassification, or temporal filtering as the exact cause.

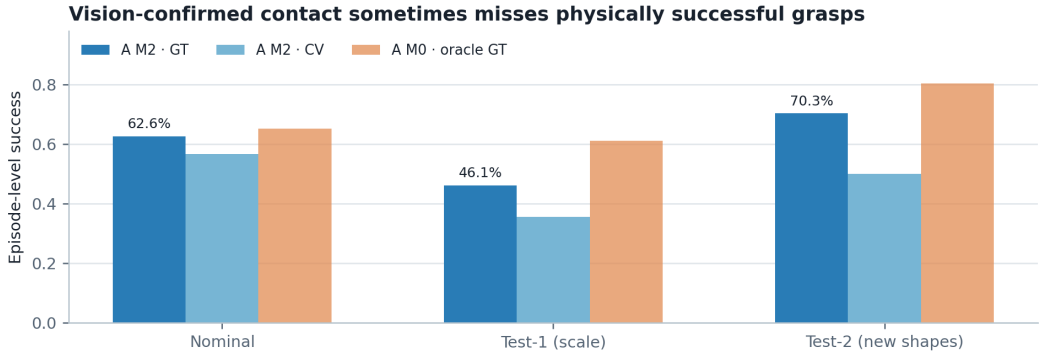


Figure 5: GT versus CV success for the camera-confirmed SAC agent. The oracle M0 is included as a separate physical-success reference; the two contact signals must not be conflated.

The independent nominal comparison of M2 versus Blind is $62.6\% - 65.6\% = -3.0$ percentage points in GT success, while M0 exceeds M2 by 2.6 points. No consistent three-seed improvement from presenting D2 contact features to the actor was demonstrated. In Test-1, M0 exceeded M2 by 15.0 points, but the study contains only three seeds and should not be reported as a statistically established method ranking.

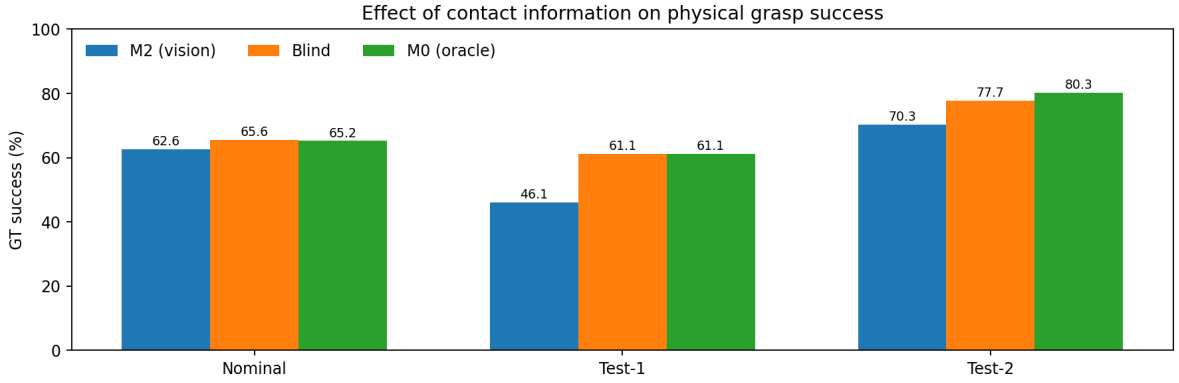


Figure 6: Contact-information ablation and feedback sensitivity in the final experimental results. The plot supplied by the V3.3 postprocessing pipeline is preferred when available; the portable fallback visualizes the GT success rates from Table 2.

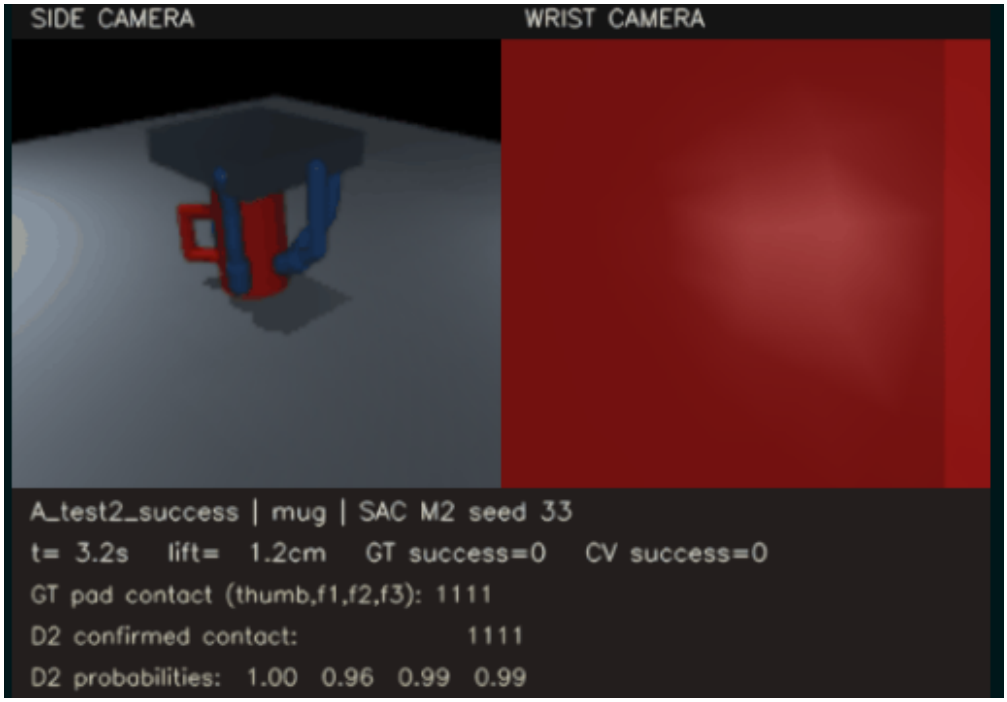


Figure 7: Frame from the Level-A SAC M2 seed-33 demonstration on an unseen mug (Test-2). At $t = 3.2$ s, the on-screen indicators are GT success = 0 and CV success = 0, with a lift of about 1.2 cm. These are *intermediate frame values*, not a statement about the full episode. The selected video is labeled as an eventual GT-success/CV-failure example (GT = 1, CV = 0 at episode level).

8 Failure Analysis, Limitations, and Discussion

The result tables distinguish never lifting, lifting then dropping, insufficient confirmed fingertip opposition, and success retained to the end. In nominal Level-A M2 evaluation, 19.8% of episodes never lifted the object, 11.3% lifted then dropped, and 6.3% held an object without satisfying the required ground-truth pad-contact pair. On Level B, 39.3% never lifted at all. These categories show why a nonzero lift height is not an adequate grasp-success metric by itself.

A short qualitative video set was captured from the stored policies. The Test-1 ring screenshot in Figure 8 provides a concrete example of a lift-height failure with contact agreement. Figure 7 depicts an *intermediate* frame from an unseen-object replay; only the full replay establishes its eventual GT/CV outcome. Figure 9 shows the simulated environment. The full MP4 recordings are retained under `visuograsp_full_outputs/cv_v2/drl_runs_v3_final/videos/v33_final_demos/`.

Several limitations remain important. First, all learned results come from SAC; the full TD3/PPO algorithm comparison requested by the broader project was not completed. Second, only three training seeds and 100k steps per run were feasible, compared with the specification's larger suggested budget. Third, the object-generalization experiment covers five unseen *procedural* families but not the requested physical YCB Test-3 set. Fourth, the Level-B policy retains a scripted trajectory prior and should not be described as full autonomous approach–grasp–lift control. Fifth, the NoSlip solver setting requires a separate physical-validity or sensitivity study before making real-world transfer claims.

Finally, three random seeds do not support strong claims of statistical significance for small differences among M0, M2, and Blind. The estimates involve many test episodes per seed but only three independent training runs. Training-checkpoint maxima are particularly optimistic and should not replace independent evaluation. The most defensible claims are descriptive: successful learning above random Level-A behavior, reduced critic-value instability, measurable GT/CV disagreement, and the absence of a demonstrated advantage over competent scripted control.

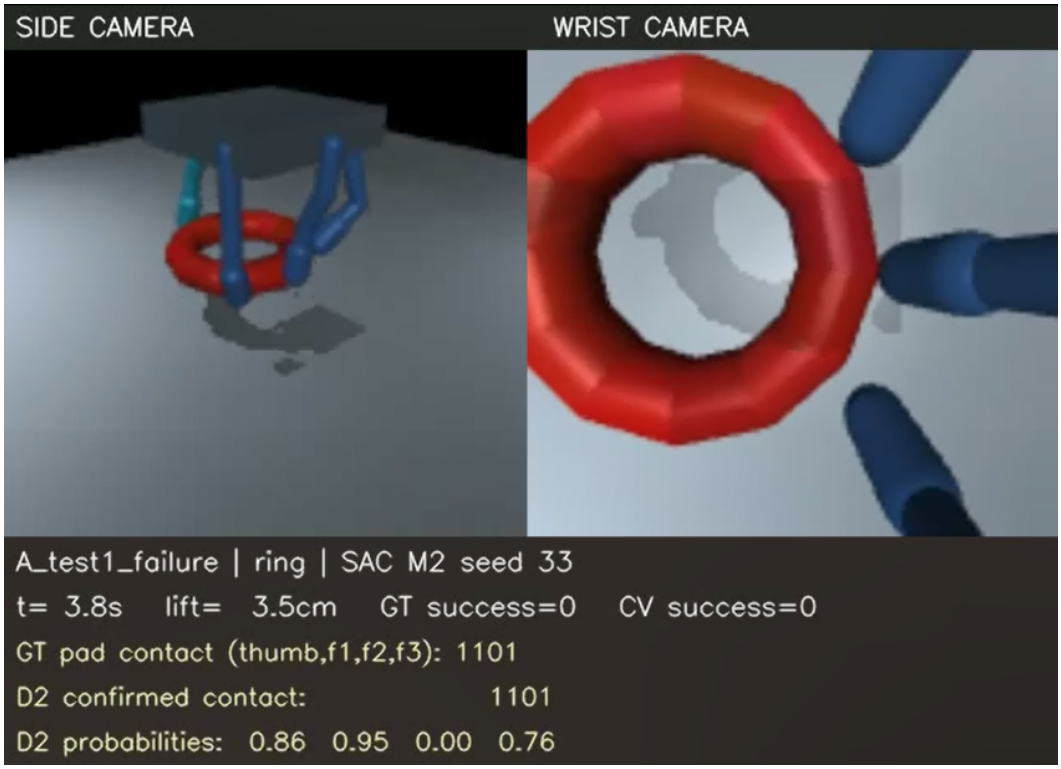


Figure 8: Qualitative failure example: Level-A SAC M2, seed 33, ring object from Test-1. At $t = 3.8$ s the ring has been raised only 3.5 cm, below the 10 cm threshold, so GT and CV success are both zero. The displayed ground-truth and D2 contact bit vectors agree (1101) at this frame: the visible failure is insufficient lift, *not* a contact-detector disagreement.



Figure 9: Representative MuJoCo scene and object geometry. Scene views illustrate the simulation environment, not the terminal success of an individual rollout.

9 Conclusion

VisuoGrasp combines MuJoCo multi-finger physics, dual-view RGB-D perception, temporal contact confirmation, and a continuous-action SAC policy. After diagnosing contact-solver issues and unstable one-step critic learning, the final LayerNorm/ten-step SAC configuration completed 12 training runs without the earlier runaway critic values. Independent Level-A M2 success reached 62.6% nominal and 70.3% on unseen procedural families, exceeding a random-action baseline but falling short of the suggested 70% nominal target and a strong scripted controller. Level B did not demonstrate superiority to manual steering. Vision-confirmed contact remained an important experimental constraint rather than a proven performance advantage: CV success lagged physical success on new shapes and M2 did not consistently outperform Blind. The central result is therefore a functional and reproducible benchmark that reveals both learnable grasping and the current limits of camera-confirmed closed-loop control.

A Reproducibility Notes

Image provenance. Figures showing screenshots of the mug and ring are frames supplied from the recorded demonstration runs. Their instantaneous overlays must not be read as final episode labels. The original final V3.3 visualizations take precedence over the portable fallback charts included with this LaTeX package.

The final notebook is `VisuoGrasp_V3_ACTIVE_v33.ipynb`. The output tree is `visuograsp_full_outputs/cv_v2/drl_runs_v3_final/`. The `models/` directory contains selected and final checkpoints; `logs/` contains training evaluations and Q-health CSVs; `final_eval/` stores episode-level independent results; and `results/v3_report/` stores the summary tables and figures. This report uses the independently evaluated values shared after the full V3.3 evaluation, not the maximum seen during training.

Reproduction caveat. The notebook's evaluation uses nominal episode indices 10–39, Test-1 indices 0–9, and Test-2 indices 0–19 for each object family and seed. A new presentation video should replay and verify the selected episode rather than infer success from a single initial frame. The exact dataset and video directories must be retained for full artifact reproducibility. The two replay frames in this report are `figures/test2_mug_frame.png` and `figures/test1_ring_failure.png`. The three final V3.3 plots are `fig1_learning_curves.png`, `fig2_success_by_condition.png`, and `fig3_where_feedback_matters.png` under `results/v3_report/`; the LaTeX source looks for these original plots before using portable fallback figures.

A.1 Selected checkpoint and evaluation protocol

The model-selection rule uses the training-shape validation set rather than the final test conditions. Table 3 preserves which snapshots were actually submitted to the final evaluator. The three selection seeds identify *independent policies*; the many episodes within each seed do not constitute additional independent training runs.

Configuration	Seed 11	Seed 22	Seed 33	Mean selection GT
Level A / SAC M2	25k / 63.9%	50k / 69.4%	75k / 75.0%	69.4%
Level A / SAC M0	50k / 75.0%	25k / 58.3%	25k / 63.9%	65.7%
Level A / SAC Blind	100k / 72.2%	25k / 75.0%	50k / 61.1%	69.4%
Level B / SAC M2	50k / 38.9%	25k / 61.1%	75k / 55.6%	51.9%

Table 3: Checkpoint step and observed selection success. These values were *not* used as the principal outcome claims.

A.2 Outcome categories on the nominal test

The independent evaluator records the physical drop flag explicitly, instead of equating early episode termination with dropping. In Table 4, *never lifted* means insufficient height, *lifted then dropped* means the drop flag fired, and *held without pad pair* means the object reached the height criterion without satisfying the required thumb-plus-finger contact pair.

A.3 Visual contact across motion phases

Frame-level D2 diagnostics aggregate confirmed contact predictions across the evaluated M2 and baseline trajectories. The nominal frame counts differ across phases and fingering configurations. The recall values below are descriptive: they should not be interpreted as success rates for complete episodes.

Nominal outcome rate	A M2	A M0	A Blind	B M2
Never lifted	19.8%	22.8%	19.8%	39.3%
Lifted, then dropped	11.3%	6.5%	10.4%	3.3%
Held without valid pad pair	6.3%	5.6%	4.3%	5.7%
Success retained to end	60.6%	63.7%	63.7%	50.9%

Table 4: Selected nominal outcome categories; rates need not sum to 100%, since categories such as success later lost are omitted.

Finger	Closing recall	Lift recall	Top recall	Top precision
0 (thumb)	73.3%	94.8%	97.1%	94.5%
1	65.7%	83.6%	85.3%	92.0%
2	74.7%	90.3%	90.8%	92.4%
3	66.5%	84.1%	88.4%	92.5%

Table 5: Aggregate D2 confirmation statistics across trajectory phases.

References

[1] E. Todorov, T. Erez, and Y. Tassa, "MuJoCo: A physics engine for model-based control," *IROS*, 2012.

[2] T. Haarnoja, A. Zhou, P. Abbeel, and S. Levine, "Soft Actor-Critic: Off-policy maximum entropy deep reinforcement learning with a stochastic actor," *ICML*, 2018.

[3] S. Fujimoto, H. van Hoof, and D. Meger, "Addressing function approximation error in actor-critic methods," *ICML*, 2018.

[4] *VisuoGrasp: Learning to Grasp Diverse Shapes with Deep RL and Vision-Confirmed Contact*, project instructions supplied with the coursework, sections 4–11.